Role of Scyther and AVISPA in IoV–Blockchain Security Analysis

- **Why Formal Verification is Needed in IoV Blockchain?**

In IoV blockchain systems, vehicles exchange **safety-critical data** over open wireless channels. Even if blockchain provides immutability and decentralization, **communication protocols themselves may still be vulnerable**.

AVISPA and Scyther are used to **formally verify** that the proposed authentication and key agreement protocol is **cryptographically secure** before deployment.

- **Role of Scyther in IoV Blockchain**

Scyther is used to analyze the scalability and robustness **of the same protocol under** unbounded IoV environments**.**

- **What Scyther Verifies**

For IoV blockchain systems, Scyther checks:

- Security properties across **infinite vehicle sessions**
- Correct agreement on session keys and identities
- Absence of replay and synchronization attacks
- Proper ordering of protocol messages

- **Why Scyther is Important for IoV**

- IoV networks are **large-scale and dynamic**
- Scyther validates that security holds even with **thousands of vehicles**
- Faster analysis compared to model-checking tools
- Ideal for **IoT/IoV lightweight protocols.**

Step-by-Step: How Scyther Is Used in IoV Blockchain

- **Step 1: Write Role-Based Protocol Description**

  - Define roles: Vehicle, RSU
  - Model exchanged messages symbolically
  - Blockchain assumed as trusted verification source

- **Step 2: Declare Cryptographic Elements**

  - Nonces
  - Session keys
  - Hash and encryption operations

- **Step 3: Define Security Claims**

Add claims such as:

  - Secret(session_key)
  - Alive(Vehicle)
  - Agreement(Vehicle, RSU, session_key)
  - Synchronization(Vehicle, RSU)

- **Step 4: Execute Scyther Analysis**

  - Tool automatically checks protocol
  - Considers **unbounded number of sessions**

- **Step 5: Interpret Results**

- No attacks → Protocol is scalable and secure
- Attack found → Tool provides counterexample trace

- **Role of AVISPA in IoV Blockchain**

AVISPA is used to **formally validate the logical correctness** of the IoV–blockchain authentication protocol under a **strong attacker model**.

- **What AVISPA Verifies**

In an IoV blockchain context, AVISPA ensures:

- Session keys exchanged between **Vehicle** ↔ **RSU** remain secret
- Mutual authentication is correctly achieved
- Replay, impersonation, and MITM attacks are prevented
- Protocol logic is secure under **bounded execution sessions**

- **How AVISPA Is Used in IoV Blockchain**

Step 1: Define Protocol Participants

Identify roles:

- Vehicle (V)
- Roadside Unit (RSU)
- Trusted Authority (TA)

Blockchain is abstracted as a **trusted immutable storage**.

Step 2: Specify Protocol in HLPSL

- Define each role separately
- Model message exchanges (nonces, timestamps, keys)
- Use cryptographic primitives (hash, encryption)
- Represent blockchain access as secure shared data

Step 3: Define Security Goals

Specify goals such as:

- secrecy_of session_key
- authentication_on vehicle_rsu
- authentication_on rsu_vehicle

Step 4: Run AVISPA Backends

Execute verification using:

- OFMC (on-the-fly model checking)
- CL-AtSe (constraint solving)

Step 5: Analyze Output

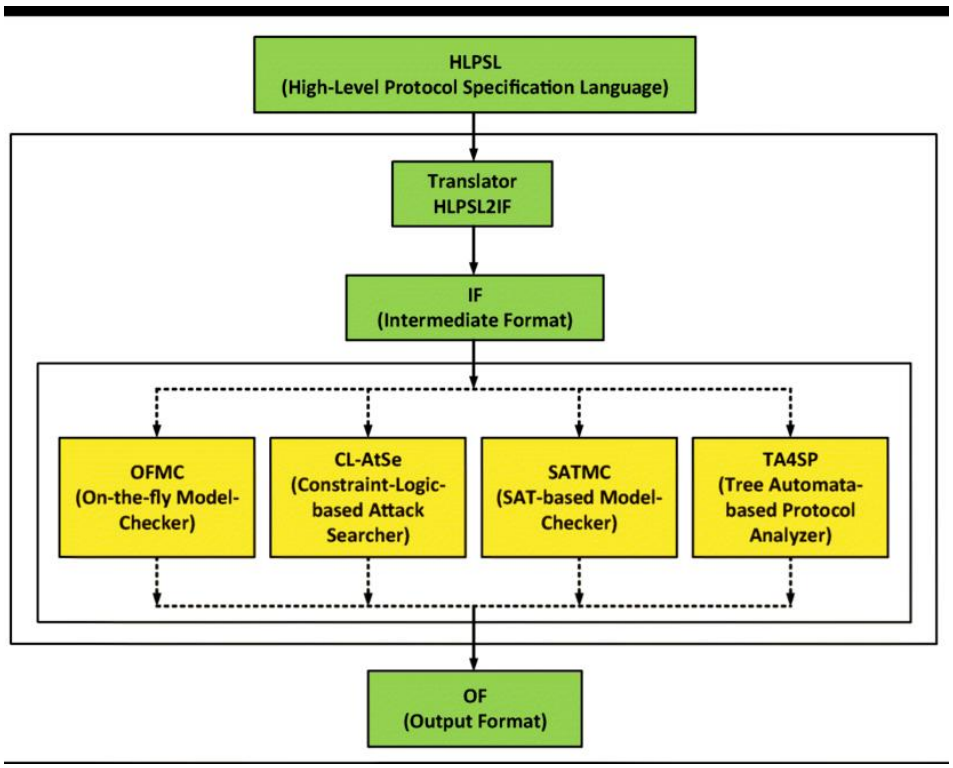
- **SAFE** → Protocol is secure
- **UNSAFE** → Attack trace reveals vulnerability

- **How AVISPA and Scyther Complement Each Other**

Aspect	**AVISPA**	**Scyther**
Verification Scope	Bounded sessions	Unbounded sessions
Strength	Deep logical correctness	Scalability & efficiency
Attack Detection	Detailed attack traces	Fast symbolic analysis
Best Use	Protocol design validation	Large-scale IoV validation

- **Comparative Summary**

Aspect	**AVISPA**	**Scyther**
Specification Language	HLPSL	Role-based protocol language
Session Handling	Bounded sessions	Unbounded sessions
Analysis Technique	Model checking & constraint solving	Symbolic verification
Attack Traces	Text-based	Graphical + textual
Complexity Handling	High (but computationally heavy)	Lightweight and fast
Typical Use Cases	Internet security, standard protocols	IoT, VANETs, lightweight protocols



Scyther results : verify

Claim				Status	Comments
enhanced	R	enhanced,R1	Secret IDS	Ok	No attacks within bounds.
		enhanced,R2	Secret Kh	Ok	No attacks within bounds.
		enhanced,R3	Secret sk(R)	Ok	No attacks within bounds.
		enhanced,R4	Niagree	Ok	No attacks within bounds.
		enhanced,R5	Alive	Ok	No attacks within bounds.
		enhanced,R6	Weakagree	Ok	No attacks within bounds.
		enhanced,R7	Nisynch	Ok	No attacks within bounds.
	T	enhanced,T1	Secret IDS	Ok	No attacks within bounds.
		enhanced,T2	Secret Kh	Ok	No attacks within bounds.
		enhanced,T3	Secret sk(R)	Ok	No attacks within bounds.
		enhanced,T4	Secret n2	Ok	No attacks within bounds.
		enhanced,T5	Niagree	Ok	No attacks within bounds.
		enhanced,T6	Nisynch	Ok	No attacks within bounds.
		enhanced,T7	Alive	Ok	No attacks within bounds.
		enhanced,T8	Weakagree	Ok	No attacks within bounds.

Done.